

Quantum-assisted Optimization of Parkinson's Deep Brain Stimulation (DBS) Parameters

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Abstract

Parkinson’s disease (PD) is associated with pathological beta-band (13–30 Hz) oscillations in basal ganglia circuits, and clinically effective deep brain stimulation (DBS) often suppresses this abnormal activity. Yet DBS programming remains a difficult optimization problem involving multiple constrained, patient-specific parameters within limited trials. Our contribution is turning that search into a rigorous benchmarked optimization problem.

We developed a simulation-based benchmarking framework to test whether *quantum-assisted optimization* can identify effective DBS settings under realistic trial budgets. Using a Parkinsonian *STN–GPe oscillation model*, we defined normalized beta-band power as the primary objective, computed *brute-force ground-truth optima* over large discretized search spaces, and scored all methods by regret under identical evaluation budgets. Our quantum-assisted pipeline fits a quadratic surrogate to simulator evaluations, converts it into a *Quadratic Unconstrained Binary Optimization (QUBO)* problem, and applies the *Quantum Approximate Optimization Algorithm (QAOA)* to propose candidate settings.

To our knowledge, this is the *first rigorous ground-truth benchmark* applying quantum-assisted optimization to Parkinson’s DBS tuning. In smooth low-dimensional tuning, classical CMA-ES performed best overall. Under clinically motivated constraints, QAOA became increasingly competitive. As the problem became more constrained and combinatorial, *QAOA moved from non-leading to competitive to best-performing*. Beyond demonstrating measurable quantum competitiveness, this work establishes a new benchmark framework for quantum neuroengineering and identifies the discrete, constraint-heavy DBS regimes in which quantum optimization may matter most.

Introduction

Parkinson’s disease (PD) is associated with pathological beta-band (13–30 Hz) oscillations in basal ganglia circuits, and clinically effective deep brain stimulation (DBS) often suppresses this abnormal activity. Because elevated beta synchronization correlates with motor impairment and is reduced by therapy, beta-band power provides a useful quantitative biomarker for Parkinsonian network dysfunction.

Despite its clinical effectiveness, DBS programming remains difficult. Therapeutic outcomes depend on multiple interacting variables—including stimulation frequency, amplitude, pulse width, and contact configuration—while clinicians can test only a limited number of settings per session. In practice, DBS tuning is therefore a constrained optimization problem in which symptom improvement must be balanced against safety limits, side effects, and energy cost.

Classical optimization methods can reduce search burden, but they remain strongest in smooth, low-dimensional parameter spaces and do not fully resolve the broader challenge of multi-parameter, clinically constrained DBS tuning. At the same time, quantum optimization has emerged as a promising framework for discrete, structured, and constrained search problems. However, most quantum applications in Parkinson’s disease have focused on diagnosis or classification rather than therapy tuning, leaving a clear gap between quantum optimization theory and the practical demands of DBS programming.

This paper addresses that gap by introducing a simulation-based, ground-truth benchmark for Parkinsonian DBS tuning under realistic evaluation limits. Using an STN–GPe oscillation model and normalized beta-band power as the primary objective, we compare strong classical optimizers against a quantum-assisted pipeline based on surrogate modeling, QUBO

construction, and QAOA. We then increase clinical realism in successive stages by adding contact geometry, charge-density feasibility, impedance-weighted energy costs, and finally a fully combinatorial contact-selection benchmark.

This is the first rigorous ground-truth benchmark applying quantum-assisted optimization to Parkinson’s DBS tuning under evaluation-budget realism. Our results show a clear progression: classical methods dominate smooth low-dimensional tuning; QAOA becomes competitive when clinically motivated constraints are added; and in a direct combinatorial contact-selection benchmark, QAOA achieves the lowest mean regret among the tested methods. Rather than assuming quantum superiority in advance, this study identifies the specific DBS tuning regimes in which quantum-assisted optimization becomes most relevant.

Review of Literature and Related Work

Pathological beta-band synchronization is one of the strongest systems-level signatures of Parkinson’s disease and an important target for therapy. Wang et al. (2018), *Pallidal Deep-Brain Stimulation Disrupts Pallidal Beta Oscillations and Coherence with Primary Motor Cortex in Parkinson’s Disease*, showed that therapeutic DBS disrupts exaggerated beta activity and pathological network coherence in Parkinson’s patients. This supports the use of beta-band power as a clinically meaningful biomarker for Parkinsonian dysfunction and DBS response.

A second line of work shows that DBS programming can be framed as an optimization problem. Louie et al. (2021), *Semi-Automated Approaches to Optimize Deep Brain Stimulation Parameters in Parkinson’s Disease*, demonstrated that classical optimization methods such as Bayesian optimization can reduce the burden of parameter search. Their work establishes that algorithmic tuning is feasible and clinically relevant, but also highlights an important limitation:

most existing approaches still operate in relatively narrow tuning regimes rather than addressing the full constrained, multi-parameter nature of DBS programming.

Quantum methods have also begun to appear in Parkinson's-related research, although mainly in diagnosis rather than treatment optimization. Vatsavai, Iyer, and Nair (2025), *A Quantum Inspired Machine Learning Approach for Multimodal Parkinson's Disease Screening*, applied quantum-inspired machine learning to multimodal Parkinson's classification. More broadly, Ngo et al. (2025), *QOMIC: Quantum Optimization for Motif Identification*, showed that quantum optimization can be applied to structured biological search problems, including neurodegenerative disease networks. Together, these studies make quantum methods plausible in biomedicine, but they do not address DBS parameter tuning.

The gap addressed by the present work is therefore specific: while the literature provides (1) a validated Parkinsonian biomarker, (2) evidence that DBS tuning can be optimized algorithmically, and (3) examples of quantum methods in diagnosis or biological network analysis, *there is no rigorous ground-truth benchmark for quantum-assisted optimization in Parkinsonian DBS tuning*. That is the space this study is designed to fill.

Problem Statement and Research Question

Deep brain stimulation (DBS) programming in Parkinson's disease is a constrained optimization problem, not a simple matter of increasing stimulation until symptoms improve. Programmable variables such as stimulation frequency, amplitude, pulse width, and contact configuration interact with physiological, safety, and hardware constraints. As a result, improving one dimension of treatment can worsen another: stronger stimulation may suppress pathological beta activity more effectively, but may also increase charge density, energy demand,

or side-effect risk. The relevant objective is therefore not an unconstrained maximum, but a *therapeutic window*.

This difficulty is compounded by limited clinical testing time. Even when the discretized parameter space contains thousands or tens of thousands of candidate settings, only a small subset can realistically be evaluated during a programming session. The practical problem is therefore not simply to identify the best setting in theory, but to find a *near-optimal setting efficiently under a limited evaluation budget*.

A further complication is structural. When DBS tuning is restricted to a few relatively smooth continuous parameters, strong classical optimizers such as CMA-ES may be expected to perform well. If quantum-assisted optimization is to become meaningfully relevant in DBS programming, it should therefore be tested in settings where the problem becomes more *constrained, discrete, and interaction-heavy*. This observation motivated the staged benchmark developed in this study: a baseline tuning problem, followed by clinically motivated constraints, and finally a direct combinatorial contact-selection benchmark.

The central research question is:

Under realistic trial budgets, can a quantum-assisted optimizer achieve competitive regret relative to strong classical methods in identifying DBS settings that suppress pathological beta oscillations?

A secondary question follows:

Does the relative performance of quantum-assisted optimization improve as the DBS tuning problem becomes more clinically realistic, constrained, and structurally combinatorial?

From these questions, we derive the working hypothesis of the study:

Under limited evaluation budgets, a quantum-assisted optimization pipeline will achieve measurable, competitive performance relative to strong classical optimizers, and its competitiveness will improve as clinically motivated constraints increase the structural complexity of the tuning problem.

The aim of this research is not to assume quantum superiority in advance, but to determine *under what conditions*, if any, it begins to emerge. In that sense, the benchmark itself is part of the contribution: it makes the problem measurable, comparable, and testable rather than merely speculative.

Methodology

Benchmark design

We designed this study as a simulation-based optimization benchmark for Parkinsonian DBS tuning. All methods were evaluated on the same Parkinsonian simulator, the same objective, and the same trial budgets, with brute-force ground truth used as the reference standard. The goal was not only to find low-cost DBS settings, but to compare how efficiently different optimizers approach the true optimum under clinically realistic evaluation limits.

The benchmark was organized into three stages.

- Stage 1 considered baseline tuning over the standard DBS parameters: frequency, amplitude, and pulse width.
- Stage 2 added clinically motivated realism through contact mode, charge-density feasibility, and impedance-weighted energy cost.
- Stage 3 isolated a direct combinatorial contact-selection problem, transforming the optimization into a QUBO-native binary search space.

Performance in all stages was evaluated by *regret relative to brute-force ground truth*.

Parkinsonian Oscillation Model

We modeled pathological beta oscillations using a simplified *STN–GPe dynamical system*, chosen because recurrent delayed feedback in this loop can sustain oscillations in the Parkinsonian beta range. DBS was represented as a periodic pulse train injected into the STN input channel. The model was not intended to reproduce the full biology of Parkinson’s disease, but to provide a controlled testbed that generates stable Parkinsonian-like beta oscillations and responds measurably to stimulation changes. The final calibrated baseline oscillation frequency was approximately *19.53 Hz*.

Objective Function

For each simulated DBS setting, the model produced a time-series signal derived from the difference between STN and GPe activity after removal of an initial transient. Oscillatory strength was quantified using Welch power spectral density estimation, and the primary efficacy metric was defined as the normalized beta-band power

$$\text{BetaRatio} = \frac{\text{BetaPower}(\text{DBS})}{\text{BetaPower}(\text{Baseline})}$$

A value near 1 indicates little or no suppression of pathological beta activity, while lower values indicate stronger suppression. This objective was chosen because beta-band activity is a reproducible systems-level biomarker of Parkinsonian dysfunction and DBS response.

Stage 1: Baseline DBS Tuning Benchmark

Stage 1 optimized three DBS parameters: *frequency*, *amplitude*, and *pulse width*. The search space was discretized into a $64 \times 64 \times 16$ grid, yielding *65,536 candidate settings*. For every point on the grid, we simulated the STN–GPe model and recorded the resulting BetaRatio. To avoid the trivial solution of simply maximizing stimulation, we constructed a Pareto front over efficacy and a normalized energy proxy, and selected a *knee point* to define a scalar benchmark objective. This preserved full ground truth while introducing a clinically meaningful efficacy–cost tradeoff.

Stage 2: Clinically Motivated Realism

Stage 2 extended the baseline problem by adding a discrete contact mode variable with two values: ring and directional/segmented. This variable modified the benchmark through three effects:

1. Charge-density feasibility: a normalized charge-density proxy penalized or excluded settings whose effective charge density exceeded a threshold.
2. Impedance-weighted energy cost: directional contacts carried a higher impedance factor, increasing the energy cost of stimulation.
3. Directional focus effect: because the simulator did not explicitly model electric fields, directional contacts were assigned a modest target-engagement advantage through amplitude scaling.

With contact mode included, the Stage 2 search space expanded to $64 \times 64 \times 16 \times 2 = 131,072$ candidate settings, of which 77,504 remained feasible after charge-density filtering.

Stage 3: Combinatorial Contact Selection

To test whether quantum-assisted optimization becomes more relevant when the decision structure itself becomes combinatorial, Stage 3 reformulated DBS tuning as a *binary contact-selection problem*. We modeled *16 possible contacts*, each of which could be either active or inactive, yielding a search space of $2^{16} = 65,536$ possible contact masks. Unlike the earlier stages, we did not optimize frequency, amplitude, or pulse width in Stage 3; these temporal stimulation variables were held fixed in order to isolate the effect of combinatorial contact selection.

Each contact was assigned heterogeneous properties representing focus, risk, impedance, and pairwise overlap. To keep the benchmark connected to the STN–GPe simulator, we first fit a quadratic efficacy curve relating normalized beta suppression to *effective stimulation amplitude*. The Stage 3 objective then combined:

- an efficacy term from this fitted beta-suppression curve,
- a sparsity penalty,
- an impedance-based energy penalty,
- a risk penalty,
- and a pairwise overlap penalty.

Because all decision variables were binary and all interactions were at most quadratic, the full Stage 3 objective was directly expressible in *Quadratic Unconstrained Binary Optimization (QUBO)* form.

Ground Truth and Scoring

In Stages 1 and 2, ground truth was obtained by brute-force evaluation of the full discretized grid. In Stage 2, the scalar optimum was defined relative to a Pareto-derived knee

point over efficacy and energy. In Stage 3, the full binary search space of 65,536 contact masks was directly enumerable, so the exact optimum was obtained by exhaustive evaluation of every mask.

Optimizer performance was scored by *regret*,

$$Regret = BestFoundCost - GroundTruthCost,$$

where lower values indicate closer agreement with the true optimum.

Optimization Methods

All methods were evaluated under identical trial budgets. We compared:

- Random search
- Genetic Algorithm (GA)
- Bayesian Optimization (BO)
- CMA-ES
- Surrogate $\rightarrow$ QUBO $\rightarrow$ exact solve
- Surrogate $\rightarrow$ QUBO $\rightarrow$ QAOA

The quantum-assisted pipeline sampled simulator evaluations, fit a quadratic surrogate, converted it into a QUBO objective, and applied QAOA to propose candidate settings.

Importantly, *QAOA did not directly simulate neural dynamics*; the STN–GPe model remained entirely classical, and the quantum component operated only in the optimization layer.

Evaluation Budgets

To reflect clinically realistic programming limits, all methods were tested under fixed budgets of $B = 10, 20, 30, 40$ evaluations. Each benchmark was repeated across multiple random

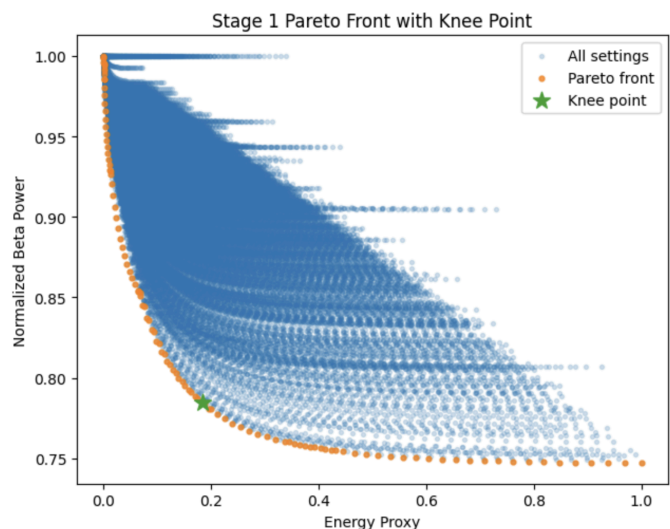
seeds, and performance was reported as *mean regret* $\pm$ *standard deviation*, allowing comparison of both accuracy and reliability.

Results

Stage 1 Baseline DBS Tuning

Stage 1 established a baseline benchmark over the three standard DBS parameters: *frequency*, *amplitude*, and *pulse width*. The calibrated STN–GPe model produced a baseline oscillation peak of approximately 19.53 Hz , confirming Parkinsonian beta-range dynamics. Brute-force evaluation over a $64 \times 64 \times 16 = 65,536$ -point grid generated the full oracle landscape and identified the ground-truth optimum under a Pareto-derived scalar objective. The Pareto front contained 109 nondominated settings, and the selected knee point occurred at approximately $f = 176.19\text{ Hz}$, $A = 0.667$, $PW = 227.33\text{ }\mu\text{s}$, with $\text{cost}^* = 0.7844$.

Figure 3. Made by participant with python. Stage 1 Pareto front showing the tradeoff between normalized beta power and stimulation-energy proxy across 65,536 settings. The green star marks the selected knee point, the benchmark optimum used for regret scoring.



To understand the geometry of the search space more directly, we also examined a cost slice through the landscape at an amplitude near the knee point. The resulting heatmap showed a relatively smooth reduction in cost as

frequency increased and pulse width approached the therapeutic region, rather than a highly fragmented or chaotic surface.

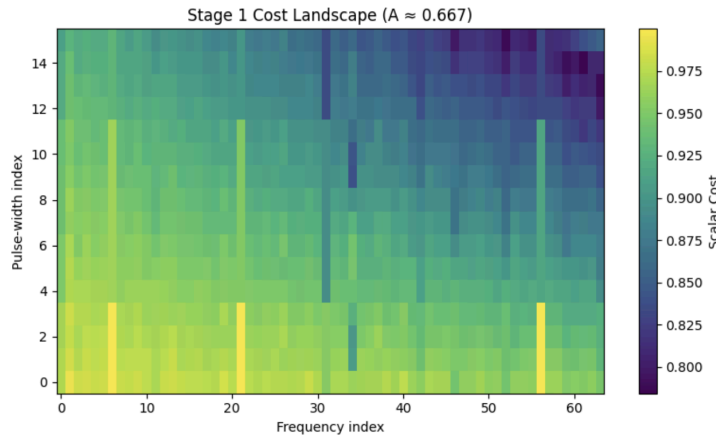


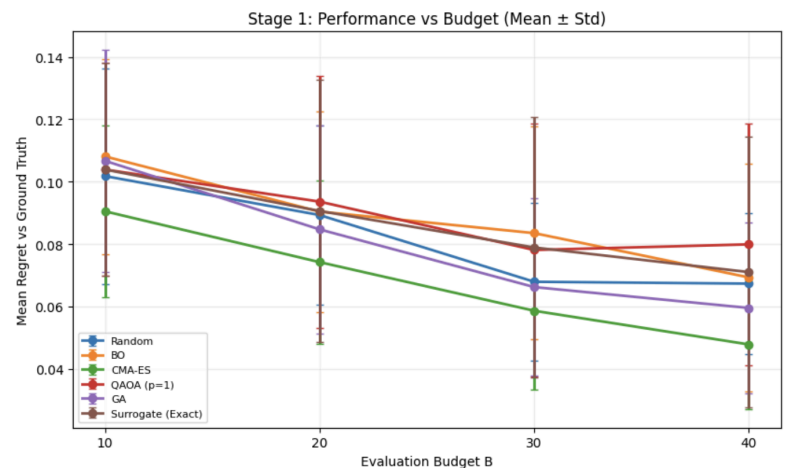
Figure 4. Made by participant with python. Stage 1 scalar cost landscape shown as a frequency-pulse-width slice at amplitude $A \approx 0.667$. Lower values indicate settings closer to the ground-truth optimum.

Figure 4 helps explain the optimizer rankings observed later. The Stage 1 landscape is broad, smooth, and only moderately nonlinear.

Method	B = 10	B = 20	B = 30	B = 40
Random	0.1018 $\pm$ 0.0346	0.0893 $\pm$ 0.0286	0.0679 $\pm$ 0.0254	0.0673 $\pm$ 0.0227
GA	0.1067 $\pm$ 0.0357	0.0847 $\pm$ 0.0334	0.0662 $\pm$ 0.0284	0.0595 $\pm$ 0.0273
BO	0.1081 $\pm$ 0.0313	0.0904 $\pm$ 0.0321	0.0835 $\pm$ 0.0341	0.0693 $\pm$ 0.0365
CMA-ES	0.0905 $\pm$ 0.0276	0.0742 $\pm$ 0.0263	0.0586 $\pm$ 0.0252	0.0478 $\pm$ 0.0209
Surrogate (Exact)	0.1039 $\pm$ 0.0342	0.0906 $\pm$ 0.0420	0.0789 $\pm$ 0.0417	0.0710 $\pm$ 0.0434
QAOA	0.1039 $\pm$ 0.0342	0.0936 $\pm$ 0.0404	0.0781 $\pm$ 0.0406	0.0799 $\pm$ 0.0387

Table 1: Made by participant with Google Docs. Stage 1 optimizer performance across evaluation budgets. Values are mean regret $\pm$ standard deviation across repeated runs. Lower regret indicates closer agreement with the ground-truth optimum.

Figure 1. Made by participant with python. Stage 1 performance vs evaluation budget. Points show mean regret relative to ground truth; error bars show ± 1 standard deviation across



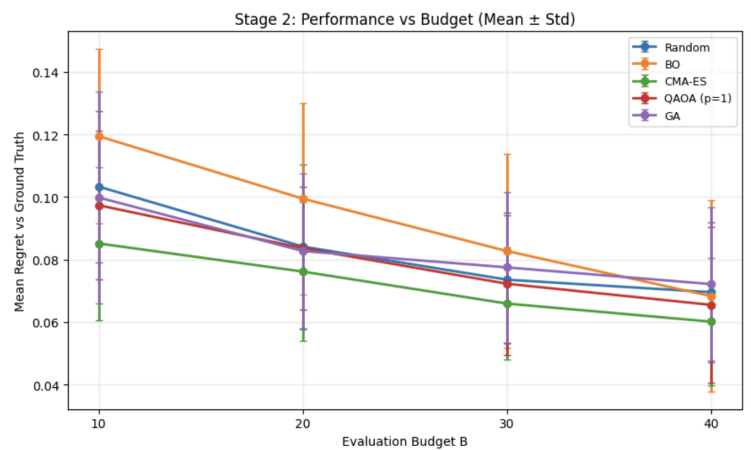
repeated runs. Lower values indicate better performance.

Across budgets $B = 10, 20, 30, 40$, *CMA-ES* consistently achieved the lowest regret, decreasing from 0.0905 ± 0.0276 at $B=10$ to 0.0478 ± 0.0209 at $B=40$. GA improved with budget and became the next strongest classical baseline at higher budgets, while BO was weaker and more variable. The QAOA pipeline improved modestly but did not outperform the best classical methods, with regret decreasing from 0.1039 ± 0.0342 at $B=10$ to 0.0799 ± 0.0387 at $B=40$. Overall, Stage 1 showed that a relatively smooth three-parameter tuning landscape strongly favors mature classical optimizers.

Stage 2 Clinically Constrained DBS Tuning

Stage 2 added contact mode, charge-density feasibility, and impedance-weighted energy cost, increasing the search space to $64 \times 64 \times 16 \times 2 = 131,072$ settings. After charge-density filtering, 77,504 settings remained feasible. The ground-truth optimum shifted to a ring-contact solution at approximately $f = 159.05$ Hz, $A = 0.5$, $PW = 240$ μ s, with $\text{cost}^* = 0.8229$.

Figure 6. Made by participant with python. Stage 2 optimizer performance versus evaluation budget. Points show mean regret relative to the ground-truth optimum, and error bars show ± 1 standard deviation across repeated runs. Lower regret indicates closer agreement with the best feasible DBS setting.



Method	B = 10	B = 20	B = 30	B = 40
Random	0.1033 $\pm$ 0.0242	0.0841 $\pm$ 0.0263	0.0736 $\pm$ 0.0205	0.0696 $\pm$ 0.0224
GA	0.0998 $\pm$ 0.0339	0.0827 $\pm$ 0.0248	0.0775 $\pm$ 0.0240	0.0721 $\pm$ 0.0245
BO	0.1194 $\pm$ 0.0280	0.0995 $\pm$ 0.0306	0.0827 $\pm$ 0.0310	0.0683 $\pm$ 0.0306
CMA-ES	0.0852 $\pm$ 0.0245	0.0762 $\pm$ 0.0222	0.0659 $\pm$ 0.0180	0.0601 $\pm$ 0.0203
Surrogate (Exact)	0.0979 $\pm$ 0.0270	0.0948 $\pm$ 0.0207	0.0912 $\pm$ 0.0220	0.0883 $\pm$ 0.0223
QAOA (p=1)	0.0974 $\pm$ 0.0237	0.0836 $\pm$ 0.0196	0.0723 $\pm$ 0.0228	0.0655 $\pm$ 0.0248

Table 2: Made by participant with python. Stage 2 results.

Under budgets B = 10, 20, 30, 40, CMA-ES again remained best overall, achieving mean regrets of 0.0839, 0.0749, 0.0658, and 0.0602,

respectively. However, unlike Stage 1, the QAOA pipeline improved more noticeably as structure and constraints were added, reaching 0.0974, 0.0860, 0.0751, and 0.0659 over the same budgets. By B=40, QAOA had moved into the top-performing group, narrowing the gap to the strongest classical method. This stage therefore provided the first clear evidence that quantum-assisted optimization becomes more competitive as the tuning problem becomes more constrained and interaction-rich.

Stage 3 Combinatorial Contact Selection

Stage 3 reformulated DBS tuning as a *binary contact-selection problem*. Instead of optimizing smooth temporal parameters, the optimizer selected among 16 contacts, each of which could be active or inactive, producing a search space of

$$2^{16} = 65,536$$

possible masks. Each contact was assigned heterogeneous *focus*, *risk*, *impedance*, and *overlap* properties, and the full objective was written directly in QUBO form. Because the entire binary space was exactly enumerable, Stage 3 provided the cleanest ground-truth benchmark in the study.

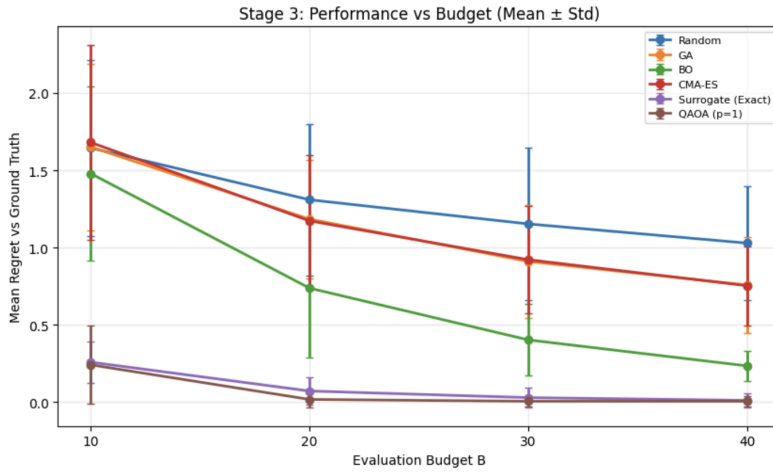


Figure 7. Made by participant with python. Stage 3 optimizer performance versus evaluation budget.

Table 3: Made by participant with python. Stage 3 optimizer performance across evaluation budgets. Values are mean regret $\pm$

Method	B = 10	B = 20	B = 30	B = 40
Random	1.6443 $\pm$ 0.5699	1.3091 $\pm$ 0.4890	1.1528 $\pm$ 0.4938	1.0286 $\pm$ 0.3656
GA	1.6504 $\pm$ 0.5397	1.1838 $\pm$ 0.3837	0.9102 $\pm$ 0.3629	0.7572 $\pm$ 0.3111
BO	1.4788 $\pm$ 0.5635	0.7375 $\pm$ 0.4514	0.4027 $\pm$ 0.2318	0.2346 $\pm$ 0.0969
CMA-ES	1.6799 $\pm$ 0.6321	1.1727 $\pm$ 0.4257	0.9213 $\pm$ 0.3448	0.7534 $\pm$ 0.2564
Surrogate (Exact)	0.2588 $\pm$ 0.1357	0.0721 $\pm$ 0.0884	0.0295 $\pm$ 0.0660	0.0116 $\pm$ 0.0433
QAOA (p=1)	0.2417 $\pm$ 0.2540	0.0178 $\pm$ 0.0534	0.0058 $\pm$ 0.0312	0.0058 $\pm$ 0.0312

standard deviation across repeated runs. Lower regret indicates closer agreement with the exact ground-truth contact-selection optimum.

Here the optimizer ranking changed dramatically. At $B = 10$, QAOA already outperformed all tested classical baselines, achieving a mean regret of 0.2417 ± 0.2540 , compared with 1.4788 ± 0.5635 for BO, 1.6443 ± 0.5699 for Random, 1.6504 ± 0.5397 for GA, and 1.6799 ± 0.6321 for CMA-ES. At $B = 20$, QAOA improved to 0.0178 ± 0.0534 , substantially outperforming the next-best method, Surrogate (Exact) at 0.0721 ± 0.0884 . At $B = 30$, QAOA again achieved the lowest mean regret (0.0058 ± 0.0312), and this value remained stable at $B = 40$, indicating near-saturation around the optimum by budget 30.

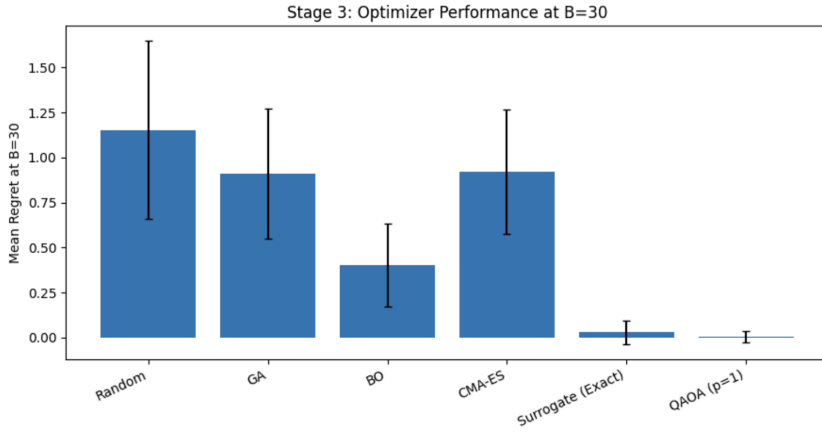


Figure 8. Made by participants using python. Stage 3 optimizer performance at $B = 30$. Bars show mean regret and error bars show ± 1 standard deviation across repeated runs.

In the Stage 3 combinatorial benchmark, QAOA reduced mean regret by approximately 75% at $B = 20$ and about 80% at $B = 30$ relative to the strongest non-quantum surrogate baseline. Stage 3 is therefore the clearest positive result in the paper. Once the DBS tuning problem was expressed as a direct combinatorial contact-selection benchmark, QAOA achieved the lowest mean regret across all tested budgets. Unlike Stages 1 and 2, where quantum-assisted optimization was non-leading or merely competitive, Stage 3 shows a regime in which QAOA becomes the strongest-performing method among those evaluated.

Discussion

The three-stage benchmark shows that the usefulness of quantum-assisted optimization in DBS tuning depends strongly on problem structure.

In Stage 1, the task was a relatively smooth, low-dimensional optimization over frequency, amplitude, and pulse width. In this regime, strong classical optimizers—especially CMA-ES—were best matched to the geometry of the search space. The cost landscape contained a broad low-cost region and only moderate nonlinearity, so efficient continuous-space search was

more important than combinatorial reasoning. This helps explain why QAOA did not outperform the strongest classical methods in the baseline setting.

In Stage 2, the benchmark became more realistic by adding contact mode, charge-density feasibility, and impedance-weighted energy cost. These additions changed the problem in two important ways: they reduced the feasible region and introduced more structure into the efficacy-cost tradeoff. As a result, the tuning problem became less like smooth knob adjustment and more like a constrained decision problem. In this regime, the QAOA pipeline improved noticeably and became competitive, even though classical CMA-ES still remained best overall. This stage therefore provides the first evidence that quantum-assisted optimization gains relevance as clinically motivated constraints are added.

The clearest result appeared in Stage 3, where DBS tuning was reformulated as a direct combinatorial contact-selection problem over $2^{16} = 65,536$ binary masks. Here the objective was naturally QUBO-native, with binary decision variables and explicit interaction terms. Unlike the earlier stages, the problem was no longer dominated by smooth continuous tuning. In this setting, QAOA achieved the lowest mean regret across all tested budgets, providing the strongest evidence in the study that quantum-assisted optimization becomes most useful when the search space is discrete, structured, and interaction-heavy.

Taken together, these results support a more precise conclusion than a generic claim of “quantum advantage.” Classical methods remain strongest in smooth low-dimensional regimes, but QAOA becomes increasingly competitive—and in Stage 3, best-performing—when DBS tuning is expressed in the kind of constrained combinatorial form that quantum optimization is designed to solve.

Limitations

This study is a *simulation-based benchmark*, not yet a clinical tool. The STN–GPe model is intentionally simplified and uses *normalized beta-band power* as a proxy for therapeutic benefit rather than direct patient outcomes such as motor scores, speech, or gait. In addition, the early stages involve relatively smooth, low-dimensional tuning problems that naturally favor strong classical optimizers. Finally, the quantum-assisted pipeline uses *QAOA as an optimization layer*, not as a direct simulator of neural dynamics. These limitations define the scope of the work: a rigorous benchmark for identifying where quantum-assisted optimization becomes competitive as DBS tuning becomes more constrained and combinatorial.

Importance and Original Contribution

This project turns DBS tuning from a difficult clinical search process into a *measurable optimization benchmark*. That is its first major impact. To our knowledge, this is the *first rigorous ground-truth benchmark* for quantum-assisted optimization in Parkinsonian DBS tuning under realistic evaluation budgets.

Its second impact is practical. DBS programming is limited by time, safety, and patient burden; an optimizer that finds better settings in the same number of trials is not just mathematically interesting, but clinically meaningful. In our results, QAOA became more competitive as the tuning problem became more constrained and, in the fully combinatorial Stage 3 benchmark, achieved the *lowest mean regret* among the tested methods.

Its third impact is conceptual. This study shows that the right question is not whether quantum computing is “better” in the abstract, but *when* it becomes better. Our benchmark identifies that turning point: as DBS tuning shifts from smooth parameter adjustment toward

discrete, structured, combinatorial decisions, quantum-assisted optimization becomes increasingly relevant. That makes this work not only an application, but a blueprint for a new direction in quantum neuroengineering.

If similar gains hold in more realistic models and future clinical systems, this could reduce programming burden by helping clinicians reach high-quality settings in fewer tested trials.

Conclusion

This study developed a simulation-based benchmark for Parkinson’s DBS tuning with ground-truth scoring and realistic evaluation budgets. Across three benchmark stages, optimizer rankings changed systematically: classical methods dominated smooth low-dimensional tuning, QAOA became competitive once clinically motivated constraints were introduced, and in the fully combinatorial, QUBO-native contact-selection benchmark, QAOA achieved the lowest mean regret among the tested methods. Rather than claiming universal quantum superiority, this work identifies the DBS tuning regimes in which quantum-assisted optimization becomes most relevant.

Future Work

- *Current steering/richer contact selection*: extend binary contact selection to multi-level current allocation across contacts.
- *Greater clinical realism*: incorporate side-effect proxies, multi-objective therapeutic-window scoring, and patient variability.
- *Closed-loop DBS*: move from static setting selection to adaptive stimulation policies using feedback.

- *Real-data validation*: compare simulator trends against open DBS/LFP datasets and published programming studies.
- *Quantum hardware testing*: evaluate QAOA on real quantum processors with error mitigation.

Bibliography

Farhi, E., Goldstone, J., & Gutmann, S. (2014). *A quantum approximate optimization algorithm*. arXiv. <https://arxiv.org/abs/1411.4028>

Louie, K. H., Petrucci, M. N., Grado, L. L., et al. (2021). Semi-automated approaches to optimize deep brain stimulation parameters in Parkinson’s disease. *Journal of NeuroEngineering and Rehabilitation*, 18, 83. <https://doi.org/10.1186/s12984-021-00873-9>

Vatsavai, D., Iyer, A., & Nair, A. A. (2025). A quantum inspired machine learning approach for multimodal Parkinson’s disease screening. *Scientific Reports*, 15, 11660. <https://doi.org/10.1038/s41598-025-95315-0>

Wang, D. D., de Hemptinne, C., Miocinovic, S., et al. (2018). Pallidal deep-brain stimulation disrupts pallidal beta oscillations and coherence with primary motor cortex in Parkinson’s disease. *Journal of Neuroscience*, 38(19), 4556–4568. <https://doi.org/10.1523/JNEUROSCI.0431-18.2018>

Chatha, K., et al. (2021). Programming deep brain stimulation for Parkinson's disease: The Toronto Western Hospital algorithms. *Frontiers in Neurology*.

Foutz, T. J., et al. (2024). A novel strategy for Bayesian optimization of deep brain stimulation therapy. *Stereotactic and Functional Neurosurgery*.

Code availability. All source code used in this study is available at:

<https://github.com/Esthercui/Parkinson-s-DBS>